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Guest Editors

Special Issue: “Artificial Intelligence Methods for Inverse Problems”

Communications in Nonlinear Science and Numerical Simulation

Prof. Davide La Torre, Prof. Herb Kunze, Prof. Oleg Michailovich, Prof. Roman Smirnov

Dear Guest Editors,

We are pleased to submit our manuscript, “*A Physics-Informed Neural Network Forward Solver for the Rough-Volatility Calibration Inverse Problem: Dimension Scaling on the Markovian-Lifted PDE and a Fast Warm-Start Hybrid*,” for consideration in the Special Issue on Artificial Intelligence Methods for Inverse Problems in *Communications in Nonlinear Science and Numerical Simulation*.

Our manuscript addresses an inverse problem that is difficult for a reason specific to this Special Issue’s scope: the forward problem itself is ill-posed before any inversion is attempted. Rough-volatility models describe market variance with a fractional, non-Markovian kernel, so there is no finite-dimensional forward PDE to solve, and consequently no forward operator to embed in a calibration inverse problem at all. We resolve this using a Markovian lift – approximating the fractional kernel by n exponentials to recover a genuine but $(n+1)$ -dimensional forward PDE – and build the resulting forward operator from a physics-informed neural network (PINN).

We believe the manuscript fits the Special Issue’s call directly on several of its named criteria:

- **Convergence.** We quantify how the AI forward operator’s own solve-error scales with the lifted dimension n (the forward problem’s governing nonlinear system), showing sub-linear growth out to $n = 32$ where a naive finite-difference alternative is already infeasible by $n = 8$.
- **Stability.** We identify four training design choices – a hard-constrained initial-condition ansatz, an integrated-variance baseline anchor, self-adaptive loss weighting, and a causal time-to-maturity curriculum – that are each individually necessary for the forward operator to converge stably, and we report the ablation showing what fails without each one.
- **Generalisation and identifiability.** We extend the forward operator to take the five model parameters as additional inputs and use it inside the calibration inverse problem itself, explicitly reporting where its accuracy generalises and where it does not (the rough, high-vol-of-vol parameter corners), and diagnosing a training-box identifiability failure that limits it as a standalone inverse-problem solver.

- **Rigorous validation against a same-fidelity reference.** Rather than validating the AI method against the true (non-lifted) model directly, we decompose its error against an independent forward solver at the same lifted dimension, isolating the network’s own error from the forward model’s approximation error – a validation methodology we believe is broadly relevant to AI methods for inverse problems built on approximated governing equations, not only to this application.

Practically, we show that the trained AI forward operator, used as a differentiable warm start for a short refinement against the true forward model, solves the calibration inverse problem for real Bitcoin option data (Deribit exchange) at approximately nine times fewer expensive forward-model evaluations than solving it from scratch, while a standalone AI-only solve is shown honestly to be insufficiently accurate on its own.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission to this Special Issue. We have no conflicts of interest to declare, and the study received no external funding.

Thank you for considering our work. We look forward to your response.

Sincerely,

Aviral Sharma